MTH304:QUIZ NO.1 **Quiz Start Time: 02:32 PM, 01 July 2022**

Question # 1 of 10 (Start time: 02:32:36 PM, 01 July 2022) **Total Marks: 1**

Greatest resultant of forces 3N and 5N is —.

Select the correct option

- 5N
- 2N
- 3N
- 8N

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 02:40:44 PM, 01 July 2022) **Total Marks: 1**

Let X be the Sum of Horizontal forces and Y be the sum of Vertical Forces acting on a particle. If X is -ve and Y is +ve , then the Resultant of the forces act in the-----

Select the correct option

- O first quadrant
- O third quadrant
- O second quadrant
- O fourth quadrant

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 02:39:59 PM, 01 July 2022) **Total Marks: 1**

If the resultant force acting on the particle is (3i+4j)N, then what additional single must be applied to make the particle in Equilibrium

Select the correct option

O	(3i-4j)N
O	(-3i-4j)N
O	(-7i-7j)N
O	(-3i+4j)N

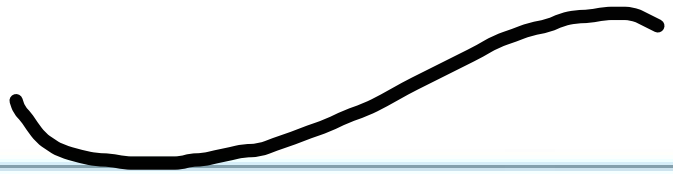
Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 02:37:42 PM, 01 July 2022) **Total Marks: 1**

If a body is moving at 2m/s, then 2m/s is the ..... of the body.

Select the correct option

O	acceleration
O	velocity
O	force
O	speed



Go to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 02:36:45 PM, 01 July 2022) **Total Marks: 1**

Lamys theorem applies to ------coplanar concurrent forces.

Select the correct option

- O two
- O four
- O three
- O five

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 02:11:27 PM, 01 July 2022)

Lamys theorem applies if a particle is in ——.

Select the correct option

○	equilibrium state
○	non-equilibrium state

Question # 2 of 10 (Start time: 02:12:07 PM, 01 July 2022)

Greatest resultant of forces 3N and 5N is —.

Select the correct option

○	3N
○	2N
○	8N
○	5N

Question # 3 of 10 (Start time: 02:13:45 PM, 01 July 2022)

The property of matter that resists changes in its motion is called....

Select the correct option

○	inertia
○	weight
○	force
○	mass

MTH304:QUIZ NO.1

Question # 4 of 10 (Start time: 02:14:13 PM, 01 July 2022)

In Lamy's theorem forces are proportional to the sine of the ------.

Select the correct option

O	sum the angle between the other two
O	difference of the angle between the other two
O	angle between the other two
O	angle adjacent to it

Question # 10 of 10 (Start time: 02:18:56 PM, 01 July 2022)

If a particle is in equilibrium under the action of coplanar Forces then removing a horizontal force makes the particle in

Select the correct option

O	neither horizontal equilibrium nor vertical equilibrium
O	vertical equilibrium only
O	horizontal equilibrium only
O	Both horizontal and vertical equilibrium



Question # 7 of 10 (Start time: 02:16:27 PM, 01 July 2022)

Let X be the Sum of Horizontal forces and Y be the sum of Vertical Forces acting on a particle then the angle that Resultant force makes with +ve horizontal is given by

Select the correct option

- $\theta = \tan^{-1}(\frac{X}{Y})$
- $\theta = \tan^{-1}(\frac{Y}{X})$
- $\theta = \tan^{-1}(YX)$
- $\theta = \tan^{-1}(Y + X)$

MTH304:QUIZ NO:1

Question # 5 of 10 (Start time: 02:14:53 PM. 01 July 2022)

In Lamy's theorem forces are.... to the sine of the angle between the other two.

Select the correct option

○	proportional
○	not proportional

Question # 6 of 10 (Start time: 02:15:32 PM, 01 July 2022)

If a body is placed on a horizontal table and the Normal component of the contact force is 100N, then the mass of the body will be ___kg

Select the correct option

○	100
○	0.1
○	1
○	10

Question # 8 of 10 (Start time: 02:17:00 PM, 01 July 2022)

Acceleration is a ..... quantity.

Select the correct option

O	both vector and scalar
O	scalar
O	vector
O	none of the given

MTH304:QUIZ NO.1

Question # 9 of 10 (Start time: 02:17:20 PM, 01 July 2022)

Law of cosines is applicable on———.

Select the correct option

- right-angled triangle only
- obtuse-angled triangle only
- any triangle
- acute-angled triangle only

BC20040885 SIDRA TUL MUNTAHA

MTH304 QUIZ NO 1

Quiz Start Time

Question # 1 of 10 (Start time: 02:20:54 PM, 01 July 2022)

If the resultant force acting on the particle is $(3i+4j)N$, then what additional single must be applied to make the particle in Equilibrium

Select the correct option

- $(3i+4j)N$
- $(3i+4j)N$
- $(3i+4j)N$
- $(7i+7j)N$

If the resultant force acting on the particle is (3i+4j)N, then what additional single must be

Select the correct option

- (-3i-4j)N
- (-3i+4j)N
- (3i-4j)N
- (-7i-7j)N

The branch of science which deals with the behavior of physical bodies when subjected to forces or displacements is called...

Select the correct option

- chemistry
- physics
- mathematics
- mechanics

Question # 3 of 10 (Start time: 02:22:31 PM, 01 July 2022)

Lamys theorem applies if a particle is in ——.

Select the correct option

○	non-equilibrium state
●	equilibrium state

Question # 4 of 10 (Start time: 02:22:56 PM, 01 July 2022)

If a particle is in equilibrium under the action of three coplanar concurrent forces, then each force has a magnitude proportional to the ----of the angle between the other two.

Select the correct option

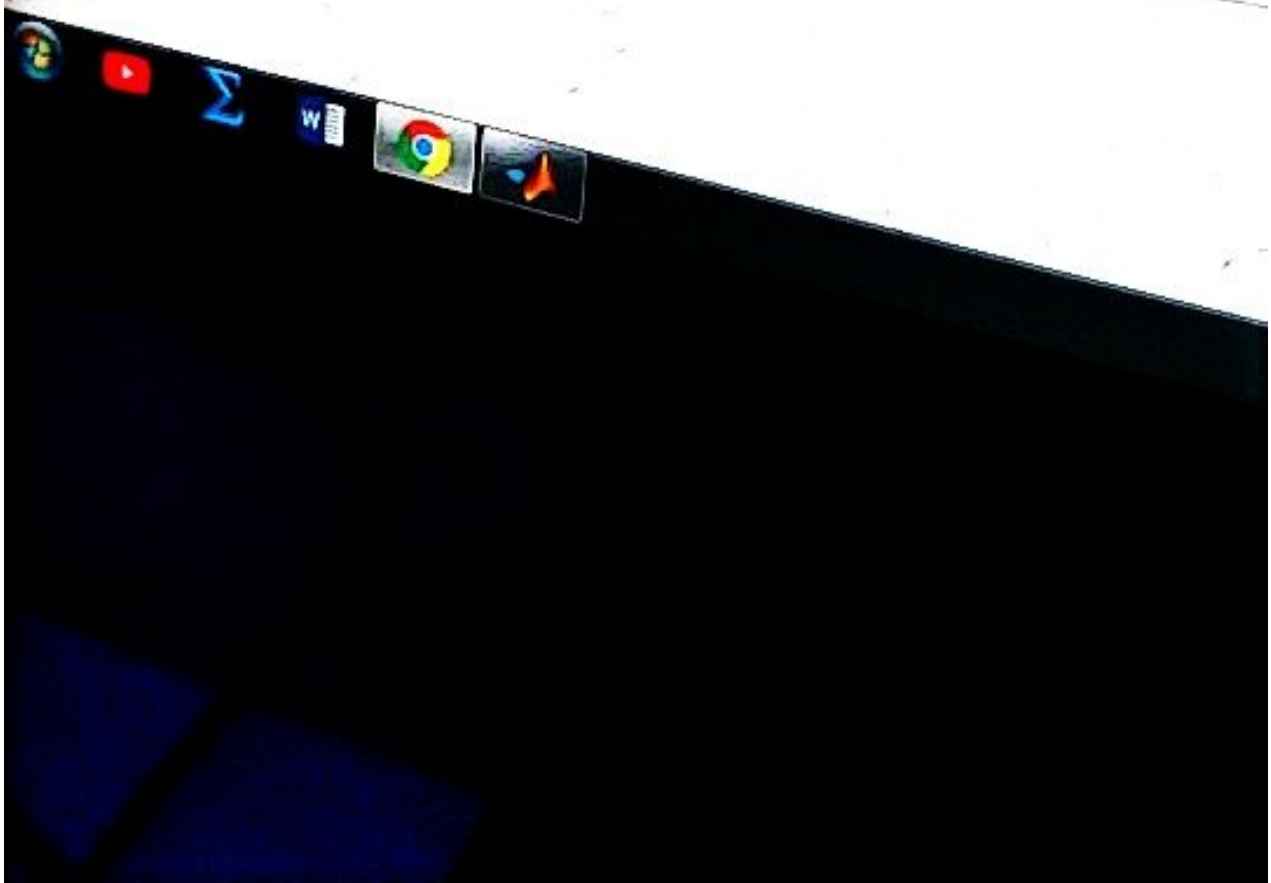
- cosine
- secant
- sine
- tangent

Question # 7 of 10 (Start time: 02:25:47 PM. 01 July 2022)

Greatest Resultant of two forces p and q is possible when angle between the two forces is—— degree

Select the correct option

- O 270
- O 90
- O 180
- O 0



Question # 6 of 10 (Start time: 02:24:52 PM, 01 July 2022)

Use of cosines is applicable on———

Select the correct option

- obtuse-angled triangle only
- right-angled triangle only
- acute-angled triangle only
- any triangle

MTH304 QUIZ NO 1

Quiz Start T

Question # 5 of 10 (Start time: 02:24:01 PM, 01 July 2022)

In Lamy's theorem forces are proportional to the sine of the ——

Select the correct option

- sum the angle between the other two
- difference of the angle between the other two
- angle adjacent to it
- angle between the other two

Let X be the Sum of Horizontal forces and Y be the sum of Vertical Forces acting on a particle then magnitude of Resultant force acting on the particle if Y=0 is given by—

Select the correct option

- O R=2X
- O R=2Y
- O R=X
- O R=Y

Question # 9 of 10 (Start time: 02:27:14 PM, 01 July 2022)

Let X be the Sum of Horizontal forces and Y be the sum of Vertical Forces acting on a particle. If X=4N and Y=0N then the angle made by the Resultant with with +ve x-axis is --

Select the correct option

- 180 degrees
- 0 degrees
- 90 degrees
- 270 degrees

Question # 9 of 10 (Start time: 02:27:14 PM, 01 July 2022)

Let X be the Sum of Horizontal forces and Y be the sum of Vertical Forces acting on a particle. If X=4N and Y=0N then the angle made by the Resultant with with +ve x-axis is ——

Select the correct option

- 180 degrees
- 0 degrees
- 90 degrees
- 270 degrees

MTH304:QUIZ NO.1

Question # 10 of 10 (Start time: 02:28:41 PM. 01 July 2022)

The property of matter that resists changes in its motion is called.....

Select the correct option

- weight
- force
- inertia
- mass

MTH304 QUIZ NO 1

Quiz Start Time: 02:32 PM, 01 July 2022

Question # 1 of 10 (Start time: 02:32:54 PM, 01 July 2022)

Total Marks: 1

Three forces (2i-j)N, (4i+3j)N and (-3i+2j)N are acting on a body. What is the magnitude of single force that must be applied on the body to make it in complete equilibrium?

Select the correct option

- 5N
- 3N
- 2N
- 4N

Question # 7 of 10 (Start time: 02:39:05 PM, 01 July 2022) **Total Marks: 1**

Let X be the Sum of Horizontal forces and Y be the sum of Vertical Forces acting on a particle then the angle that Resultant force makes with +ve horizontal is given by

Select the correct option

Reload Math Equations

O	$\theta = \tan^{-1}(\frac{Y}{X})$
O	$\theta = \tan^{-1}(\frac{X}{Y})$
O	$\theta = \tan^{-1}(Y + X)$
O	$\theta = \tan^{-1}(YX)$

Click to Save Answer & Move to Next Question

MTH304:QUIZ NO.1

Quiz Start Time: 02:32 PM, 01 July 2022

Question # 2 of 10 (Start time: 02:33:20 PM, 01 July 2022)

Total Marks: 1

Acceleration is a ____ quantity.

Select the correct option

- none of the given
- both vector and scalar
- vector
- scalar



Question # 4 of 10 (Start time: 02:35:18 PM, 01 July 2022) **Total Marks: 1**

Three forces (2i+3j)N, (4i-2j)N and (-5i+4j)N are acting on a body. What single force must be applied on the body to make it in complete equilibrium?

Select the correct option

O	i-5j
O	-i+5j
O	i+5j
O	-i-5j

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 02:41:46 PM, 01 July 2022) **Total Marks: 1**

According to Lamy's theorem

Select the correct option

- The three forces must be equal
- none of these
- If the three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two
- The three forces must be at 120° to each other

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 02:39:15 PM, 01 July 2022) Total Marks: 1

Let X be the Sum of Horizontal forces and Y be the sum of Vertical Forces acting on a particle then magnitude of Resultant force acting on the particle is given by

Select the correct option

Reload Math Equations

O	$R = \sqrt{X^2 - Y^2}$
O	$R = \sqrt{X^2 + Y^2}$
O	$R = X - Y$
O	$R = X + Y$

MTH304:QUIZ NO.1

Quiz Start Time: 02:32 PM, 01 July 2022

Question # 9 of 10 (Start time: 02:38:41 PM, 01 July 2022)

Total Marks: 1

If a body of weight 5N is hanging in equilibrium by a light inextensible string then the tension in the string is------

Select the correct option

- 0.5N
- 5N
- 0N
- 50N

Question # 8 of 10 (Start time: 02:38:16 PM, 01 July 2022) Total Marks: 1

If a body doesn't change its position with respect to its surrounding then it is said to be in state of.....

Select the correct option

- motion
- rest



Question # 7 of 10 (Start time: 02:37:11 PM, 01 July 2022) Total Marks: 1

Let X be the Sum of Horizontal forces and Y be the sum of Vertical Forces acting on a particle. If X is -ve and Y is +ve , then the Resultant of the forces act in the-----

Select the correct option

- first quadrant
- third quadrant
- fourth quadrant
- second quadrant

MTH304 QUIZ NO.1

Quiz Start Time: 02:32 PM, 01 July 2022

Question # 5 of 10 (Start time: 02:36:18 PM, 01 July 2022)

Total Marks: 1

A particle is moving in two-dimensional plane with speed 5m/s. Which of the following is NOT the possible velocity of the particle?

Select the correct option

- 3i+4j
- 2i+3j
- 5i
- 5j

Question # 3 of 10 (Start time: 02:34:18 PM, 01 July 2022) **Total Marks: 1**

While using Lamy's Theorem which of the following can be angle between any two forces?

Select the correct option

- O 80 degrees
- O 180 degrees
- O 0 degrees
- O 360 degrees

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 02:35:35 PM, 01 July 2022) Total Marks: 1

If the resultant force acts along the line y=x, then the resultant makes an angle of ---- degrees with +ve x-axis.

Select the correct option

- 45
- 90
- 135
- 0

MTH304 QUIZ NO.1

Quiz Start Time: 02:32 PM, 01 Ju

Question # 4 of 10 (Start time: 02:34:37 PM, 01 July 2022)

Total M

If a particle is in equilibrium under the action of coplanar Forces the removing a horizontal force makes the particle in---

Select the correct option

- vertical equilibrium only
- Both horizontal and vertical equilibrium
- neither horizontal equilibrium nor vertical equilibrium
- horizontal equilibrium only

MTH304 QUIZ NO 1

Quiz Start Time: 02:32 PM, 01 July

Question # 3 of 10 (Start time: 02:34:09 PM, 01 July 2022)

Total Mar

Trigonometric ratios are applicable on-----.

Select the correct option

- obtuse-angled triangle only
- any triangle
- acute-angled triangle only
- right-angled triangle only

Question # 2 of 10 (Start time: 02:33:35 PM, 01 July 2022) Total Marks: 1

Lamys theorem applies to ------coplanar concurrent forces.

Select the correct option

- four
- five
- three
- two

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Question # 2 of 5 (Start time: 07:20:46 AM,

The torque depends upon.....

Select the correct option

- moment arm only
- both force and moment arm
- mass
- force only

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MTH304 - Statics (QUIZ NO.1)

Question # 1 of 5 (Start time: 07:20:03 AM, 17 Febru

A line along which a force act is called-----.

Select the correct option

- axis of rotation
- a line of acting torque
- moment arm
- a line of action of force

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Moment of force=--------.

Select the correct option

○	F - d, where F is force and d is moment arm
○	F/d, where F is force and d is moment arm
○	F*d, where F is force and d is moment arm
○	F + d, where F is force and d is moment arm

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